Hepatitis E virus load in swine organs and tissues at slaughterhouse determined by real-time RT-PCR.

Although uncommon in North America, Hepatitis E virus (HEV) has been identified in some industrialized countries in patients without a history of travel to HEV-endemic countries. Its presence is ubiquitous worldwide in swine populations. Zoonotic transmission of swine HEV to non human primates has been achieved experimentally and transmission of HEV after ingestion of contaminated raw or undercooked meat is well documented. In Canada, so far, no HEV outbreak has been documented but HEV genotype 3 strains have been identified in sera and faecal samples of swine origin. The objective of the present study was to determine the viral load of HEV in liver, loin, bladder, hepatic lymph node, bile, tonsil, plasma and faeces samples of 43 pigs at slaughter. Feline calicivirus (FCV) was used as sample process control to validate the RNA extraction process, as a confirmation of the absence of sample inhibitors and as an amplification control. Using FCV/HEV multiplex TaqMan RT-qPCR system, HEV RNA was detected in 14 out of the 43 animals tested. HEV was detected in lymph nodes (11/43), bladder (10/43), liver (9/43), bile (8/43), faeces (6/43), tonsils (3/43), plasma (1/43) samples from infected animals. No HEV-positive loin samples were observed. Viral loads of 10(3) to 10(7) copies/g were estimated in positive liver and bile samples.